25

Random Matrices

Historical Note

John Wishart, working on multivariate statistics in 1928, wrote down the distribution of a sample covariance matrix and so studied the first random matrix. The subject proper begins with Eugene Wigner in 1955, faced with the energy levels of heavy atomic nuclei. The true Hamiltonian was unknowable and enormous, so Wigner proposed replacing it by a random symmetric matrix and asking what its eigenvalues look like, on the theory that a typical large matrix might reproduce the statistics of a particular complicated one. It did. Marchenko and Pastur found the corresponding law for sample covariance matrices in 1967. In 1972, over tea at Princeton, Hugh Montgomery described to Freeman Dyson the pair correlation of the zeros of the Riemann zeta function, and Dyson recognised it instantly as the one he had computed for random matrices. Nobody knows why.

Chapter 24 left two debts. It quoted a bound on how many rows a sketch needs in order to embed a d-dimensional subspace, and it never said where the shape $k \sim d/\varepsilon^2$ comes from. And Chapter 17 observed that elimination behaves well on matrices "that arise from applications or at random" without saying what the second half of that phrase means. Both are questions about the spectrum of a large random matrix, which is a subject with answers.

25.1 Two ensembles, and what the trace already knows

Definition 25.1 (Wigner and Wishart ensembles)**.** *A* Wigner matrix *is a symmetric $A \in M_n(\mathbb{R})$ whose entries on and above the diagonal are independent with mean 0 and variance 1. A* Wishart *or* sample covariance *matrix is $W = \frac{1}{m} X X^t$, where $X \in \mathbb{R}^{n \times m}$ has independent entries of mean 0 and variance 1; by Proposition 15.12 it is positive semidefinite.*

Before quoting anything, notice how much the trace settles on its own.

Proposition 25.2 (The scaling is forced). *For a Wigner matrix A,*

$$\mathsf{E}\,\operatorname{trace}(A) = 0, \qquad \mathsf{E}\,\operatorname{trace}(A^2) = n^2.$$

Consequently the eigenvalues $\lambda_1, \dots, \lambda_n$ of $M = A/\sqrt{n}$ satisfy

$$\mathsf{E}\Big[\tfrac{1}{n}\sum_i \lambda_i\Big] = 0, \qquad \mathsf{E}\Big[\tfrac{1}{n}\sum_i \lambda_i^2\Big] = 1.$$

Proof. The first is immediate since every diagonal entry has mean 0. For the second, A is symmetric so $\operatorname{trace}(A^2) = \sum_{i,j} a_{ij}a_{ji} = \sum_{i,j} a_{ij}^2$, a sum of n^2 terms each of expectation 1. Now $\operatorname{trace}(M) = \operatorname{trace}(A)/\sqrt{n}$ and $\operatorname{trace}(M^2) = \operatorname{trace}(A^2)/n$, and by Theorem 9.12 the trace of a matrix is the sum of its eigenvalues while the trace of its square is the sum of their squares. Dividing by n gives the two averages. □

Key Idea

That is worth pausing on, because it fixes the one thing that must be got right before any theorem can be stated. The eigenvalues of A itself grow like $\sqrt{n}$, not like n and not like a constant, and the trace of A^2 says so without any probability beyond the variance of one entry. Divide by $\sqrt{n}$ and the average squared eigenvalue is exactly 1 at every size, so the spectrum of $A/\sqrt{n}$ can settle down to a fixed shape. Getting the normalisation from the second moment in this way is the standard first move in the subject, and it is linear algebra.

25.2 The semicircle law

Assumption 25.3 (Wigner, 1955, quoted). *Let A be a Wigner matrix and $M = A/\sqrt{n}$. As $n \to \infty$ the empirical distribution of the eigenvalues of M, that is the measure placing mass $1/n$ at each λ_i, converges to the* semicircle law *with density*

$$\rho(x) = \frac{1}{2\pi}\sqrt{4 - x^2} \qquad \text{on } [-2, 2],$$

and zero outside. Its odd moments vanish and its $2k$th moment is the Catalan number $C_k = \binom{2k}{k}/(k+1)$.

Figure 25.1 puts a histogram of eigenvalues against that density. The striking feature is what the answer does not depend on: the distribution of the individual entries enters

only through its variance.

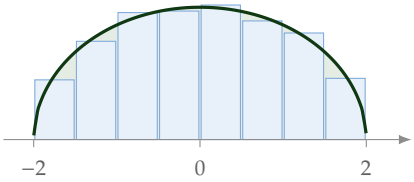


Figure 25.1. Assumption 25.3. The bars are a histogram of eigenvalues and the curve is the limiting density $\rho(x) = \frac{1}{2\pi}\sqrt{4-x^2}$. The striking feature is what the answer does not depend on: the distribution of the individual entries enters only through its variance, so a matrix of Gaussians and a matrix of random signs have the same limiting spectrum. That is called universality, and it is the reason Wigner could say something about a nucleus whose Hamiltonian he did not know. Compare Proposition 25.2: the curve has mean 0 and second moment 1, exactly as the trace demanded.

Caution

The law describes the *bulk* and says nothing about any individual eigenvalue. In particular it does not say that $\lambda_{\max} \to 2$; that is a separate theorem, true under a moment condition, and the fluctuations of $\lambda_{\max}$ around 2 are of size $n^{-2/3}$ and follow the Tracy and Widom distribution rather than a normal one. The distinction matters in applications, because it is usually the extreme eigenvalue and not the bulk that decides whether an algorithm works.

Example 25.4 (The first two moments, computed exactly)**.** *The semicircle law can be watched arriving in a computation small enough to do on paper. Take the Wigner ensemble of Definition 25.1 in its crudest form: A is $n \times n$ and symmetric, the diagonal is zero, and each entry above it is an independent fair coin, ± 1. Put $W = A/\sqrt{n}$, the scaling that Proposition 25.2 forces.*

The second moment needs no probability at all:

$$\operatorname{trace}(A^2) \;=\; \sum_{i,j} A_{ij}A_{ji} \;=\; \sum_{i,j} A_{ij}^2 \;=\; n(n-1),$$

because every off-diagonal entry squares to 1 whatever the coin said. So $\frac{1}{n}\operatorname{trace}(W^2) = (n-1)/n$ always*, with no fluctuation whatsoever, and it tends to 1.*

The fourth moment does need probability, and it is where the combinatorics starts.

Expanding $\operatorname{trace}(A^4) = \sum A_{ij}A_{jk}A_{kl}A_{li}$*, a term survives the expectation only if every edge it uses is used an even number of times, so only closed walks of length four that retrace their steps contribute. Counting them gives*

$$\mathsf{E}\operatorname{trace}(A^4) \;=\; 2n^3 - 5n^2 + 3n, \qquad \textit{hence} \qquad \tfrac{1}{n}\,\mathsf{E}\operatorname{trace}(W^4) \;=\; 2 - \frac{5}{n} + \frac{3}{n^2}.$$

At $n = 3, 4, 5$ *the left side is* $18, 60, 140$*, which one can confirm by summing over all* $2^{n(n-1)/2}$ *sign patterns; the normalised values are* $2/3$*,* $15/16$ *and* $28/25$*, climbing towards* 2*.*
Now compare with the semicircle law itself. Its moments are $\int x^{2k}\, d\mu_{sc}(x) = C_k$*, the* k*th Catalan number, and* $C_1 = 1$*,* $C_2 = 2$*. Those are exactly the two limits just computed. The Catalan numbers count non-crossing pairings, and the walks that survived the expectation are precisely the trees traced out by such pairings: the combinatorics of the limit law is already visible in a finite matrix. This is the moment method, and it is how Wigner found the law in 1955, before anyone knew what the eigenvectors looked like.*

25.3 Marchenko and Pastur, and a debt discharged

Assumption 25.5 (Marchenko and Pastur, 1967, quoted)**.** *Let* $G \in \mathbb{R}^{k\times d}$ *have independent entries of mean* 0 *and variance* 1*, and let* $k, d \to \infty$ *with* $\gamma = d/k$ *fixed in* $(0, 1]$*. Then the singular values of* $G/\sqrt{k}$ *fill out the interval*

$$\left[\,1 - \sqrt{\gamma},\; 1 + \sqrt{\gamma}\,\right],$$

and the extreme ones converge to its endpoints.

The relevance to sketching is immediate once one observes that a Gaussian sketch, restricted to a subspace, is again Gaussian. That part is linear algebra.

Proposition 25.6 (A Gaussian sketch sees a subspace as Gaussian)**.** *Let* $Q \in \mathbb{R}^{m\times d}$ *have orthonormal columns and let* $G \in \mathbb{R}^{k\times m}$ *have independent standard normal entries. Then the rows of* GQ *are independent with mean* $\mathbf{0}$ *and covariance* $\mathbf{1}_d$*; in fact* GQ *has independent standard normal entries.*

Proof. Row i of GQ is $Q^t g_i$, where g_i is row i of G, and distinct rows are independent because the g_i are. The covariance is

$$\mathsf{E}\big[(Q^t g_i)(Q^t g_i)^t\big] \;=\; Q^t\,\mathsf{E}[g_i g_i^t]\,Q \;=\; Q^t Q \;=\; \mathbf{1}_d,$$

using $Q^tQ = \mathbf{1}_d$ from Definition 12.14. That a linear image of a Gaussian vector is Gaussian, so that the covariance determines it, is the one probabilistic fact used. □

Corollary 25.7 (Where $k \sim d/\varepsilon^2$ comes from)**.** *Let $S = G/\sqrt{k}$ be a Gaussian sketch and let U be a d-dimensional subspace with orthonormal basis Q. Then $SQ = (GQ)/\sqrt{k}$ is a $k \times d$ Gaussian matrix divided by $\sqrt{k}$, so by Assumption 25.5 its singular values fill $[1 - \sqrt{d/k},\, 1 + \sqrt{d/k}]$. By Proposition 24.6, S is an ε-subspace embedding for U precisely when that band sits inside $[1 - \varepsilon, 1 + \varepsilon]$, that is when*

$$\sqrt{d/k} \;\le\; \varepsilon, \qquad \textit{equivalently} \qquad k \;\ge\; d/\varepsilon^2.$$

Key Idea

That is Assumption 24.7 of Chapter 24, no longer quoted but explained, and the explanation says that the exponent on ε cannot be improved. The square root in $1 \pm \sqrt{d/k}$ is the whole story: to shrink the band by a factor of ten one must multiply the number of rows by a hundred. Sketching is cheap in d and expensive in accuracy, and Corollary 25.7 says why that trade is a fact about random matrices rather than a weakness of anybody's proof.

25.4 Condition numbers, and what "at random" means

Corollary 25.8 (Condition number of a random matrix)**.** *In the setting of Assumption 25.5, the condition number of G in the spectral norm satisfies*

$$\kappa(G) \;\longrightarrow\; \frac{1 + \sqrt{\gamma}}{1 - \sqrt{\gamma}}, \qquad \gamma = d/k.$$

For a tall matrix, γ small, this is close to 1. For a square matrix, $\gamma = 1$, the band is $[0, 2]$, the bound is vacuous, and in fact κ grows: Edelman showed that for a square $n \times n$ Gaussian, $\mathsf{E}[\ln \kappa] = \ln n + O(1)$.

Figure 25.2 shows the band those two statements describe. A tall matrix sits at the left, where the band pinches shut and the matrix is nearly an isometry; a square one sits at the right, where the band reaches zero and the bound says nothing.

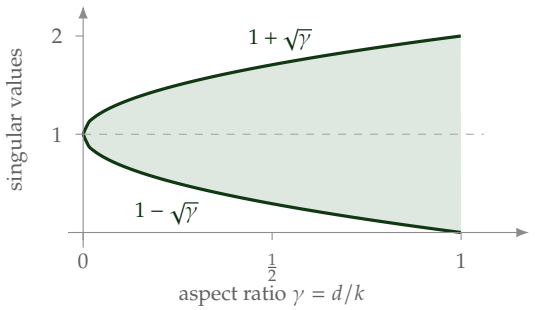


Figure 25.2. Assumption 25.5 and Corollary 25.8. Every singular value of a random $k \times d$ matrix, scaled by $\sqrt{k}$, lies in the shaded band. A tall matrix sits at the left, where the band pinches to the single value 1 and the matrix is nearly an isometry; a square matrix sits at the right, where the band reaches zero and the matrix is nearly singular. The vertical extent of the band at a given γ is $2\sqrt{\gamma}$, and that square root is what forces $k \sim d/\varepsilon^2$ in Corollary 25.7.

Insight

This is a partial answer to the puzzle left open in Chapter 17. Elimination behaves well on random matrices in the weak sense that a random matrix which is at all rectangular is well conditioned, with the conditioning degrading only as the shape approaches square, and even then only logarithmically. What it is *not* is an explanation of the growth factor. The Wilkinson example of Chapter 17 has entries ± 1 and is perfectly well conditioned, so conditioning is not the obstacle; the growth factor is a property of the elimination process rather than of the matrix, and why partial pivoting almost never triggers it remains open. Random matrix theory settles the easier half of the question and leaves the interesting half alone.

25.5 Initialisation

Proposition 25.9 (What a random layer does to a norm). *Let $W \in \mathbb{R}^{n_{out} \times n_{in}}$ have independent entries of mean 0 and variance σ^2. Then for every fixed x and g,*

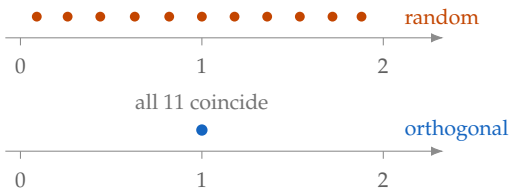
$$\mathsf{E}\,\|Wx\|^2 = n_{out}\,\sigma^2\,\|x\|^2\,, \qquad \mathsf{E}\,\left\|W^t g\right\|^2 = n_{in}\,\sigma^2\,\left\|g\right\|^2\,.$$

Proof. Both are Proposition 24.2 with the scaling restored. For the first, the n_{out} rows of W satisfy $\mathsf{E}[w_i w_i^t] = \sigma^2 \mathbf{1}$, so each term $\mathsf{E}[(w_i^t x)^2] = \sigma^2 \|x\|^2$ and there are n_{out} of them. For the second, the rows of W^t are the n_{in} columns of W, each of length n_{out}, and the

same computation applies. □

Key Idea

The two displays pull in opposite directions and that is the whole of initialisation theory. The forward pass preserves the norm of the activations when $\sigma^2 = 1/n_{\text{out}}$; the backward pass, which applies W^t by Proposition 19.4, preserves the norm of the gradient when $\sigma^2 = 1/n_{\text{in}}$. A layer that is not square cannot do both, and the familiar prescriptions are the available compromises: $1/n_{\text{in}}$ protects the gradient, $2/(n_{\text{in}} + n_{\text{out}})$ splits the difference, and the extra factor of two in the version used with a rectifier is there because that nonlinearity discards half of its input. None of this is a heuristic. It is Proposition 25.9 with a choice made about which of two incompatible goals to serve.



singular values of a square layer: spread over $[0, 2]$ if random, all equal to 1 if orthogonal

Figure 25.3. Corollary 25.8 at $\gamma = 1$. A randomly initialised square layer has singular values spread across $[0, 2]$, so it flattens some directions almost completely and doubles others; a stack of L such layers compounds that, which is the exploding and vanishing gradient problem of Example 23.12. An orthogonal initialisation puts every singular value at 1 and the problem does not arise. This is the quantitative case for the construction $Q = e^{(A-A^t)/2}$ that Chapter 23 supplied.

Caution

Everything in this chapter is a statement about matrices with independent entries, and a trained weight matrix has none of that. Empirically the spectra of trained networks depart from Marchenko and Pastur in two ways that matter: a small number of large outliers separate from the bulk, and the bulk itself often develops a heavy tail. Both are taken as signatures of learned structure rather than noise, and neither is predicted by anything above. Use these results for initialisation, where the independence assumption is true by construction, and treat their application to trained weights as description rather than theory.

Open Problem 25.10 (Do the zeros of the zeta function belong to this chapter?)**.** *The nontrivial zeros of the Riemann zeta function lie, conjecturally, on a vertical line. Write them as* $\frac{1}{2} + i\gamma_n$ *and normalise the* γ_n *to have average spacing one.* Do the normalised zeros have the same local statistics as the eigenvalues of a large random Hermitian matrix?

What is known. *Montgomery computed the pair correlation of the zeros in 1973, under the Riemann hypothesis and for a restricted range of test functions, and obtained the function* $1 - (\sin \pi u / \pi u)^2$ *[473]. Dyson recognised it instantly, over tea, as the pair correlation of eigenvalues of the Gaussian unitary ensemble, a fact from Chapter 25's own subject. Odlyzko's computations of billions of zeros high up the critical line match the random matrix prediction to striking precision. Rudnick and Sarnak extended the agreement to all n-level correlations, again for restricted test functions.* Nobody can remove the restriction, and nobody can prove the correspondence in general.

Why anyone believes it. *Katz and Sarnak proved the exact analogue over function fields, where the zeta function of a curve over a finite field really does have its zeros as eigenvalues of a Frobenius matrix, and the equidistribution of those matrices is a theorem [365]. So in the setting where the "matrix" exists, the statistics are provably random-matrix. Over the rationals, no matrix is known. Hilbert and Pólya are supposed to have suggested that one should exist, self-adjoint, with the* γ_n *as its spectrum; a century later this remains a hope rather than a programme.*

Where a reader might start. *Download the first hundred thousand zeros, normalise the spacings, and plot the histogram against the Wigner surmise* $\frac{\pi}{2} s\, e^{-\pi s^2/4}$ *and against the exponential density that independent points would give. The fit to the first and the failure of the second is visible at a glance, and it is one of the most convincing unexplained pictures in mathematics.*

Example 25.11 (Small random matrices are not small versions of large ones)**.** *Enumerate all* $2^9 = 512$ *matrices with entries* ± 1 *of size* 3×3*. Two facts fall out, and neither is what asymptotic theory would suggest.*

First, 320 of them are singular*, exactly five eighths. Singularity is not a measure-zero accident here; it is the typical case, because with only two values available the rows collide constantly. For a continuous ensemble the probability would be zero.*

Second, and more startling, every one of the 192 *nonsingular ones has condition number* exactly 2*. Not approximately: the distribution is a single atom. At* $n = 2$ *the same computation gives condition number exactly* 1 *for every nonsingular sign matrix, because such a matrix is* $\sqrt{2}$ *times an orthogonal one.*

The lesson is about what the asymptotic results of this chapter are for. Corollary 25.8

describes κ for large n, where the entries have room to arrange themselves in many ways and a limiting distribution appears. At $n = 3$ there is no room: the discreteness of the entries dominates, the answer is rigid, and no limit law is visible. Random matrix theory is a large-n subject not by preference but by necessity, *and testing its predictions on a 3×3 example will mislead every time.*

Figure 25.4 follows the support of the limiting law as the matrix becomes square. The interesting end is the left one: it reaches zero exactly when the aspect ratio does, which is the same statement as the condition number blowing up.

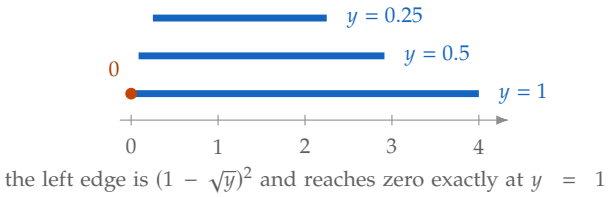


Figure 25.4. The Marchenko and Pastur support, as the matrix becomes square. For an $n \times d$ Gaussian matrix with $y = d/n$, the eigenvalues of the sample covariance fill $[(1-\sqrt{y})^2, (1+\sqrt{y})^2]$, and the interesting end is the left one. At $y = 0.25$ the support is $[0.25, 2.25]$ and the condition number is $\sqrt{2.25/0.25} = 3$; at $y = 0.5$ it is 5.83; at $y = 1$ the left edge touches zero and the condition number is unbounded. *A square random matrix is ill conditioned not by accident but by law*, and this is the quantitative form of Corollary 25.8. It is also the reason every sketching bound in Chapter 24 asks for more rows than columns: the gap $n - d$ is what holds the smallest singular value away from zero.

Exercises

Exercise 25.12. *Repeat Proposition 25.2 for entries of variance τ^2 rather than 1, and deduce the correct normalisation and the radius of the resulting semicircle. Check your answer against the case $\tau = 1$.*

Exercise 25.13. *Compute $\mathsf{E}\,\mathrm{trace}(A^4)$ for a Wigner matrix to leading order in n by counting which index patterns in $\sum_{i,j,k,l} \mathsf{E}[a_{ij}a_{jk}a_{kl}a_{li}]$ survive, and confirm that the fourth moment of $A/\sqrt{n}$ tends to 2, the Catalan number C_2.*

Exercise 25.14. *Verify directly that* $\int_{-2}^{2} x^2 \rho(x)\,dx = 1$ *and* $\int_{-2}^{2} \rho(x)\,dx = 1$ *for the semicircle density. The substitution* $x = 2\sin\theta$ *makes both routine.*

Exercise 25.15. *For* $d = 500$ *tabulate the Marchenko and Pastur band and the limiting condition number at* $k = 1000, 5000, 50000$*. How many rows are needed for* $\kappa \le 1.1$*, and how does that compare with the* $k \ge d/\varepsilon^2$ *of Corollary 25.7 at* $\varepsilon = 0.05$*?*

Exercise 25.16. *Prove Proposition 25.6 in the case* $d = 1$ *by hand, taking* Q *to be a single unit vector, and identify exactly where* $\|q\| = 1$ *is used. What would go wrong if the columns of* Q *were merely independent rather than orthonormal?*

Exercise 25.17. *Explain why Corollary 25.8 gives no information at* $\gamma = 1$*, and why that is the correct answer rather than a defect of the bound. Then reconcile it with the fact that a square Gaussian matrix is invertible with probability one.*

Exercise 25.18. *A layer has* $n_{in} = 1024$ *and* $n_{out} = 256$*. Give the variance prescribed by each of the three rules in the Key Idea after Proposition 25.9, and compute in each case the expected factor by which the forward pass and the backward pass scale a norm.*

Exercise 25.19. *Using Proposition 25.9, compute the expected squared norm of the output of a stack of* L *square layers of width* n *with entry variance* σ^2*. For which* σ *is it independent of* L*, and why does Figure 25.3 show that this is not enough to make the stack well behaved?*

Exercise 25.20. *Take* A *a random* 2×2 *matrix with independent standard normal entries, form* $Q = e^{(A - A^{t})/2}$ *as in Example 23.12, and verify by hand that* Q *is a rotation for every* A*. Which single number of* A *determines the angle?*

References and further reading

Potters and Bouchaud [515] is the right first book for a reader arriving from this one: it is written for people who want to use the results, it proves the semicircle and Marchenko and Pastur laws by methods that need no measure theory, and it is good on the applied questions of Section 25.5. Tao [598] is the elegant mathematical treatment and is where

to find the moment method that Exercise 25.13 is a first step in. Bai and Silverstein [40] is the comprehensive reference for the sample covariance case, including the versions of Assumption 25.5 with dependent or non-identically distributed entries. Edelman [206] is the original on condition numbers and is still the clearest account of what happens at $\gamma = 1$. Vershynin [627] gives the non-asymptotic bounds, which are what one actually needs to choose a sketch size at finite k rather than in the limit, and so is the natural companion to Chapter 24. Banica [51], our baseline, ends its final part with random matrices approached through the moment method and free probability, which is where this chapter stops.